

Password-Based ELECTRICAL APPLIANCE CONTROL

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In today's technologically advanced world, simplicity and security are paramount. Password-based electrical appliance control is a cost-effective solution that combines both these aspects. This project introduces a novel way to operate electrical appliances, such as lights or fans, using a secure password as the key. In this article, we delve into the concept, circuitry, working principle, and potential applications of this innovative system.

Often, mischievous individuals, especially kids, find amusement in repeatedly ringing doorbells and then hastily disappearing. This vexing behaviour is driven by two factors: the use of a single switch to activate the doorbell and the absence of any security measures. Fortunately, the concept of this circuit extends beyond just appliance control. It can also serve as a password-based doorbell, offering an extra layer of security to any home.

The password-based electrical appliance control is a straightfor-

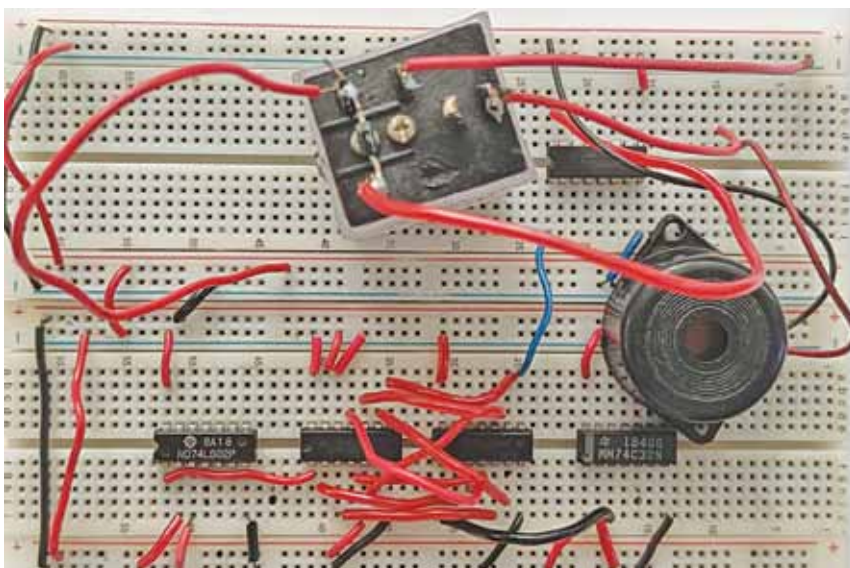


Fig. 1: Prototype tested in EFY lab

ward yet ingenious project that employs a secure password to enable the operation of electrical appliances. By implementing this automatic system, a control assembly facilitates the seamless on/off control of appliances using a password. The core components of this system include logic gates and a relay driver IC, which work in tandem to ensure the correct functioning of the setup.

Fig. 1 shows the prototype tested in EFY lab in which buzzer was used as an appliance. And we used jumper wires in place of switches.

Circuit and working

The heart of the system lies in its circuit design, depicted in Fig. 2. The circuit utilises various logic gate ICs, including OR gate 74LS32 (IC1), NAND gate 74LS00 (IC2), NOT gate 74LS02 (IC3), and NOR

gate 74LS02 (IC4). Additionally, a relay driver IC ULN2003 (IC5) and a 5V, one changeover relay (RL1) are integral to the setup. Seven on/off switches (S1 through S7) contribute to the password entry process. For practical application, an electrical appliance—here a 60W, 230V bulb—serves as the load.

To activate the load, the final output from the logic gate circuit is directed to pin 10 of IC3E. This output is then connected to pin 1 of ULN2003 IC, which subsequently interfaces with the relay to activate the load—be it a light, a buzzer, or any other appliance.

In addition to appliance control, the system can be adapted for use with a 230V AC operated doorbell. To ring the doorbell, the correct password sequence must be input using the switches. This innovation

PARTS LIST

Semiconductors:

- IC1 (IC1A-IC1E) - 74LS32 quad 2- input OR gate
- IC2 (IC2A-IC2E) - 74LS00 quad 2- input NAND gate
- IC3 (IC3A-IC3G) - 74LS04 hex inverter (NOT gate)
- IC4 (IC4A-IC4E) - 74LS02 quad 2- input NOR gate
- IC5 - ULN2003 relay driver

Miscellaneous:

- CON1-CON3 - 2-pin connector
- RL1 - 5V one changeover relay
- S1-S7 - On/off switch
- EA - 60W, 230V AC bulb